

UK PATENT APPLICATION

SYSTEM AND METHOD FOR AUTONOMOUS CROSS-DOMAIN CAUSAL DISCOVERY AND BUSINESS GENERATION USING ENSEMBLE PREDICTION AND TEST-DRIVEN DEVELOPMENT

ABSTRACT

A computer-implemented system for autonomous cross-domain causal discovery and business generation comprising: a data ingestion pipeline collecting behavioural and macroeconomic time-series data from heterogeneous sources; a correlation engine computing pairwise associations using multiple correlation coefficients; a bidirectional Granger causality module determining temporal causal direction via frequency-adaptive lag selection; a three-model ensemble prediction engine combining Granger causality, ordinary least squares regression, and autoregressive integrated moving average forecasts with automatic weight calibration from historical accuracy; a cross-layer enforcement architecture constraining predictions from behavioural signal variables to macroeconomic target variables; an exploitation scoring module ranking opportunities; an autonomous test-driven development pipeline converting statistical signals into deployed web applications; a deployment module deploying generated applications without human intervention; a commercial intelligence module aggregating deployment revenue and engagement metrics to score configuration confidence and enrich future build specifications; and a build-iterate loop diagnosing failures and spawning corrective builds.